

2.5D cQED

From Whispering Photons to Transmons

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Thanks to:

Zaki Leghtas

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Matt Reagor

Teresa Brecht

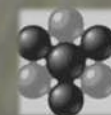
Leonid Glazman

Rob Schoelkopf

Steve Girvin

QuLab

RSL Lab



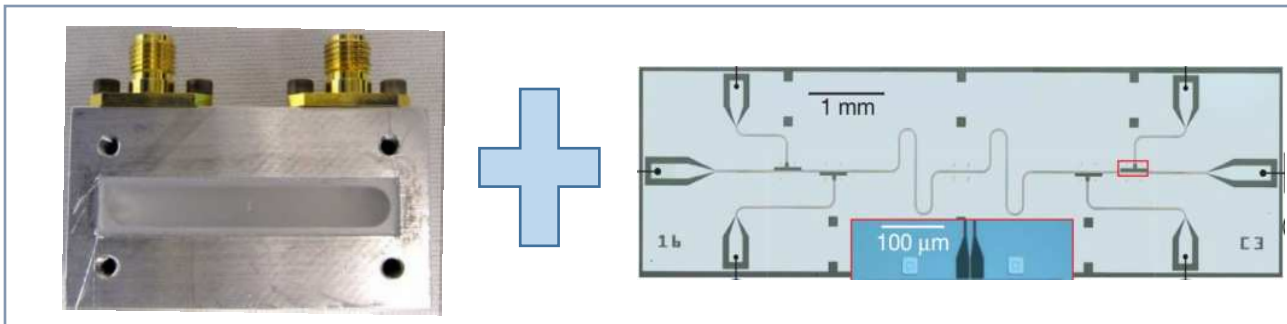
Yale Institute for Nanoscience
and Quantum Engineering

Outline

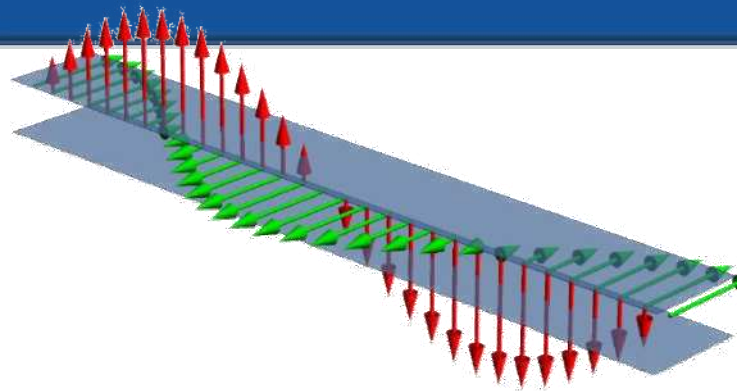
1. Whispering gallery mode resonators:
Concept & coherences
2. “Energy BBQ”
3. Integrated two cavity – one Transmon device

Motivation

- I. Marry wafer scalability & 3D quality?
 - Coupled resonators & multiple qubits
 - On chip control lines
- II. Test materials at the single photon level
 - Thin films
 - Dielectrics



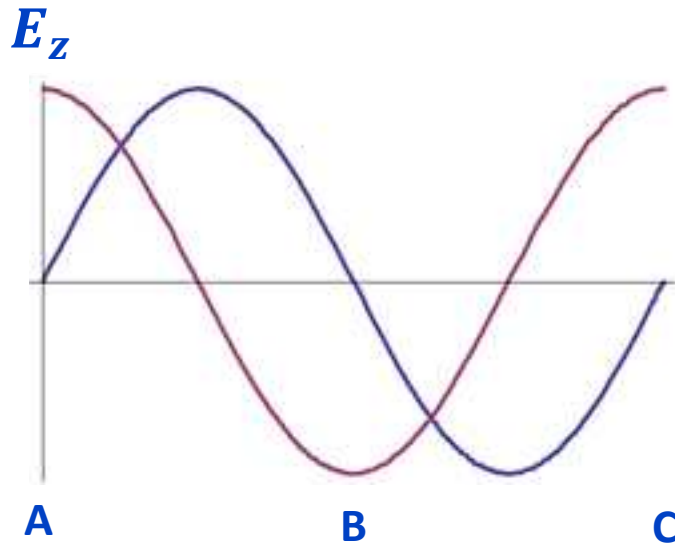
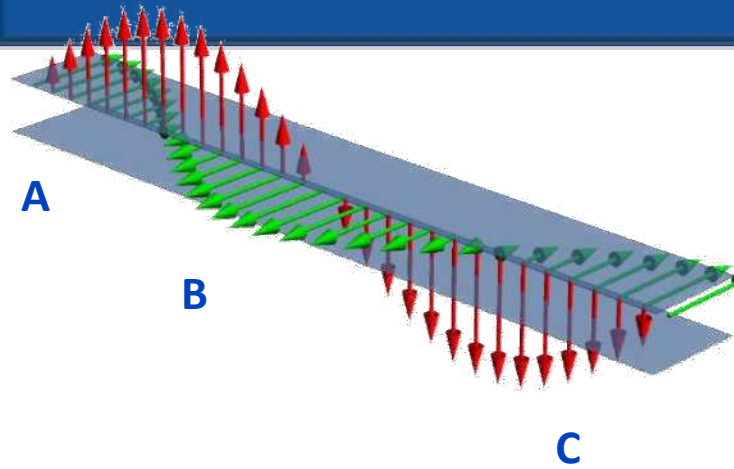
Design strategy:



- Confine energy in lossless vacuum
- No seams
- Keep geometry & design parameters simple

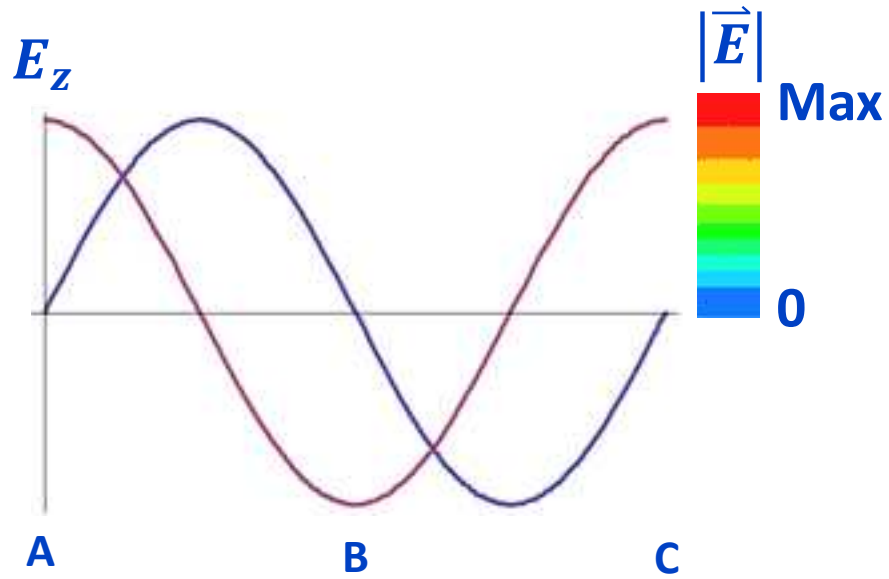
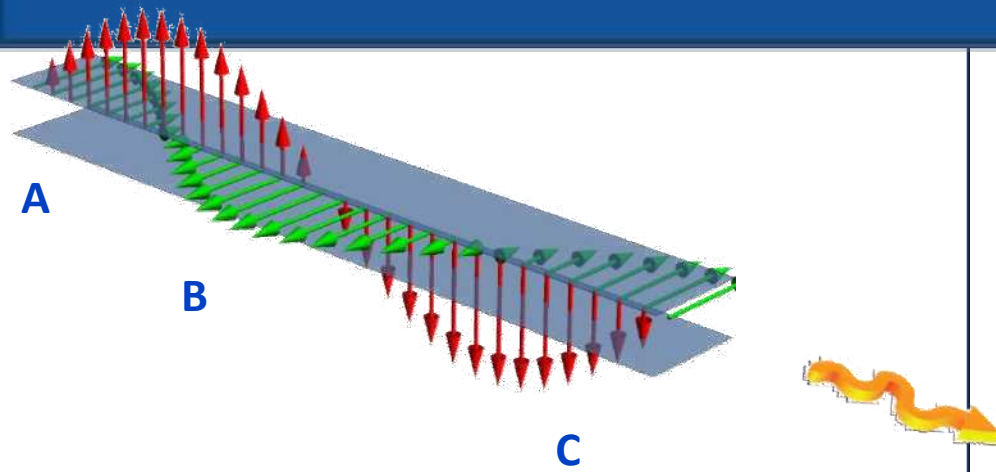
How to form a resonator

Periodic boundary conditions

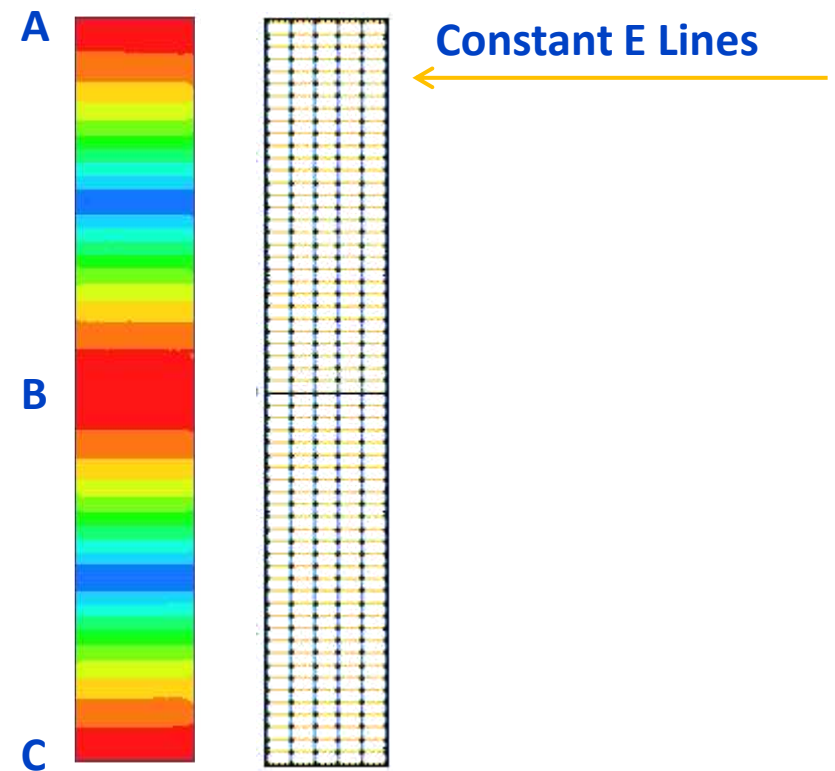


Degenerate ground state

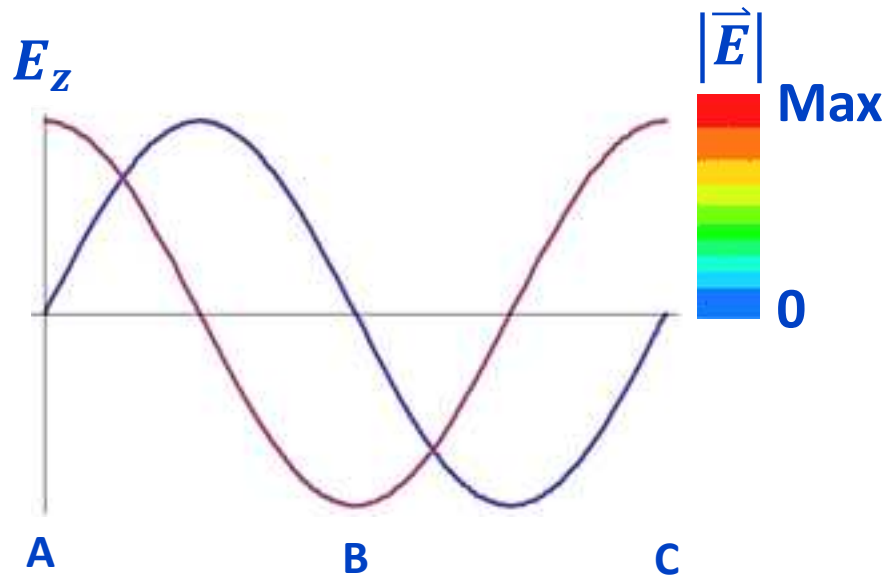
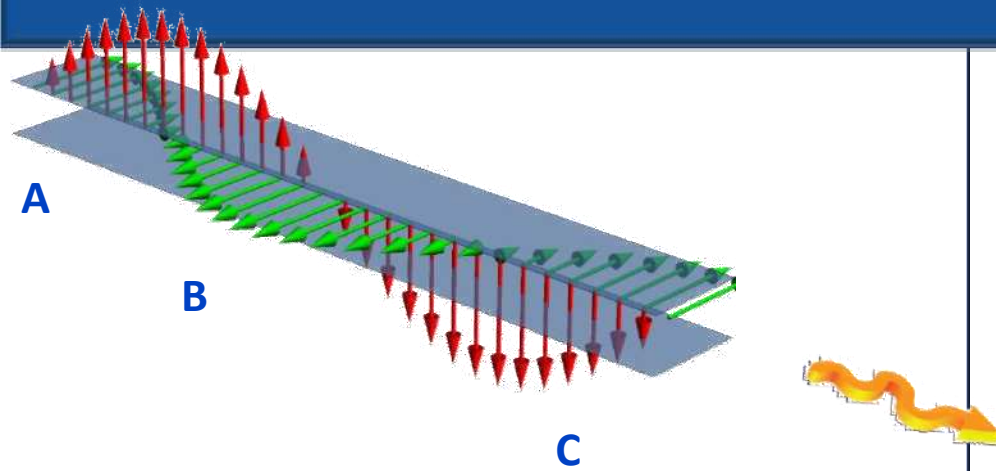
But it's `2D`: The color map



Top View ($|\vec{E}|$)



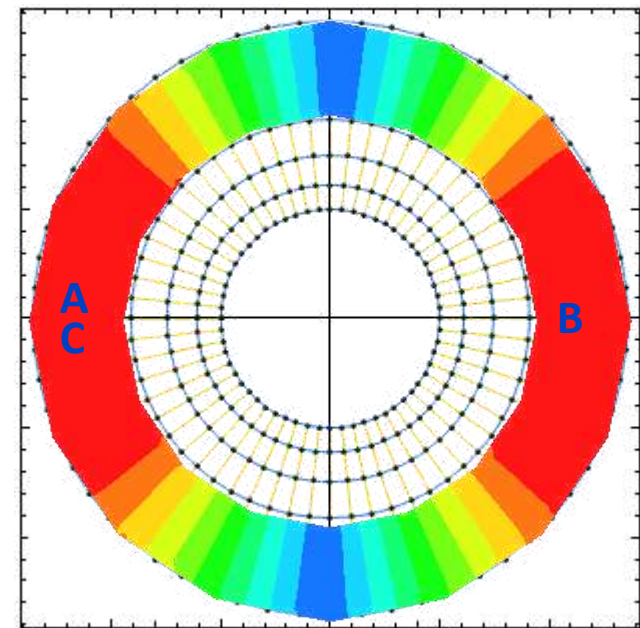
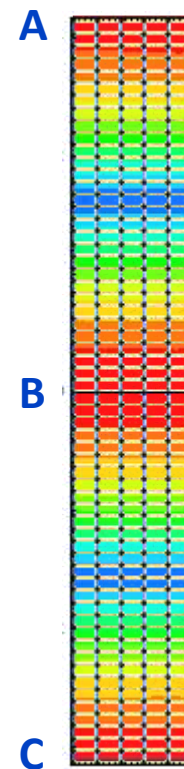
Physical periodic boundary conditions: Hello ring



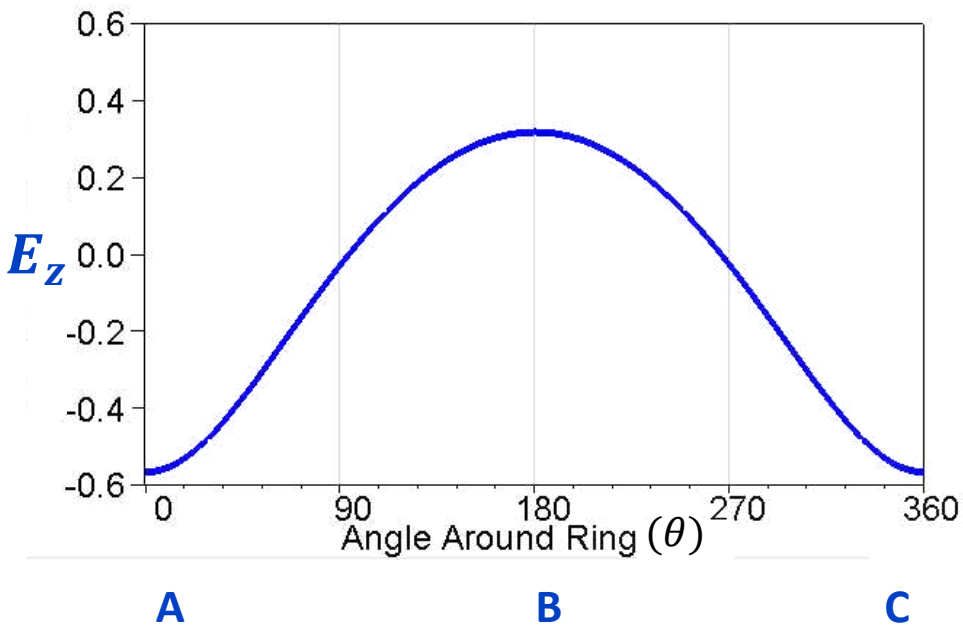
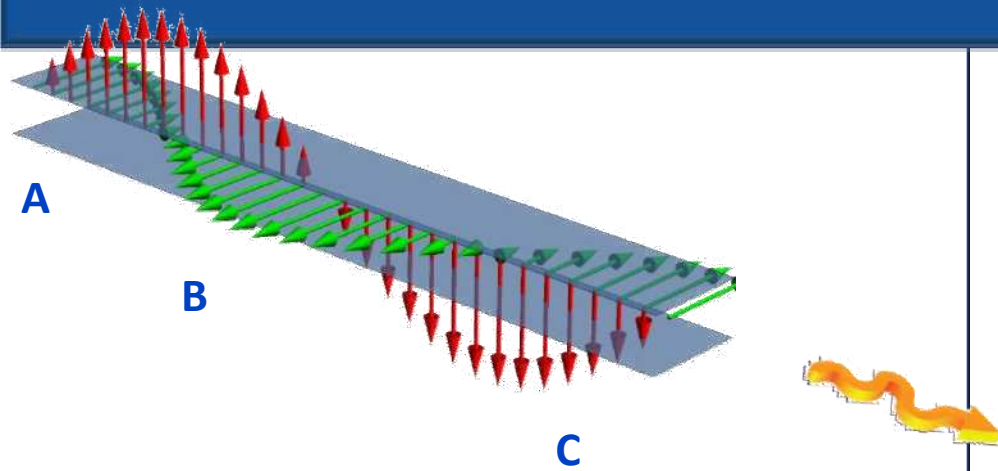
Top View ($|\vec{E}|$)

z-plane $\longrightarrow$ w-plane

$$w = e^z$$



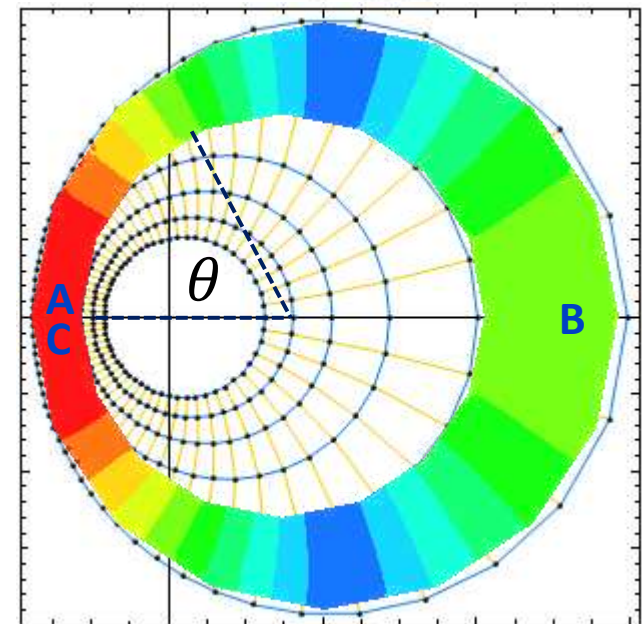
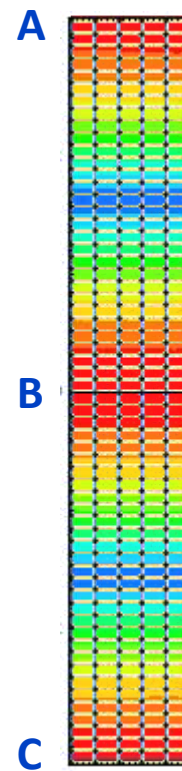
Warped Ring



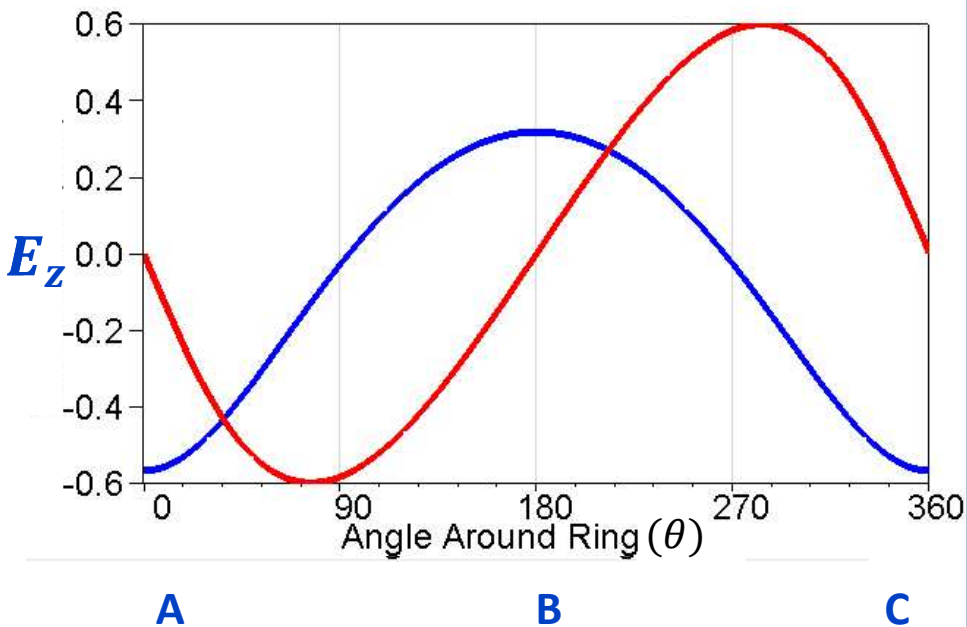
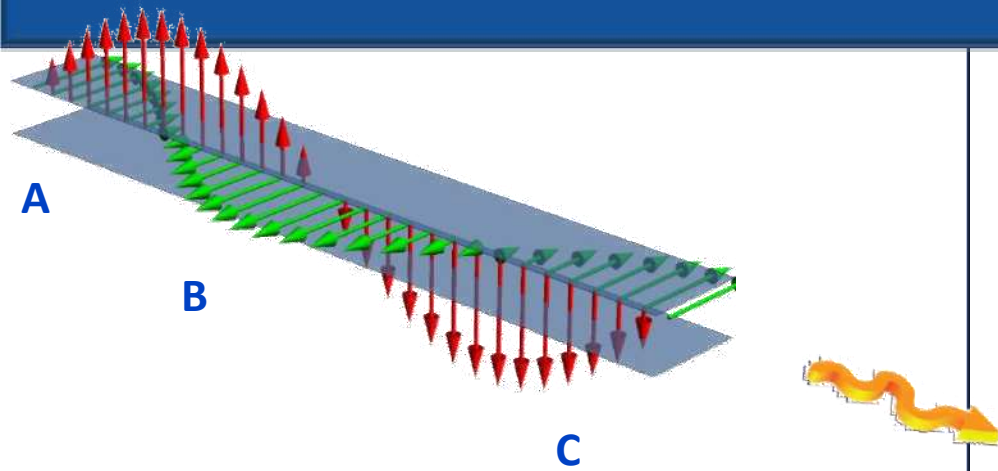
Top View ($|\vec{E}|$)

z-plane $\longrightarrow$ w-plane

$$w = \frac{e^z}{1 - s e^z}$$



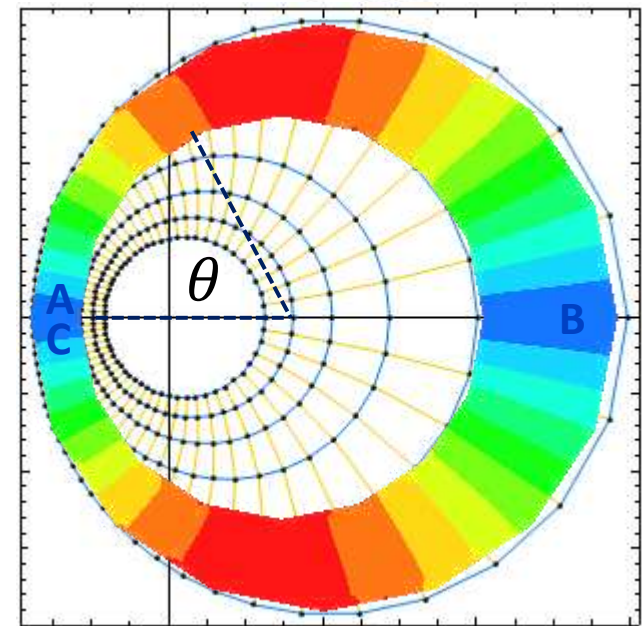
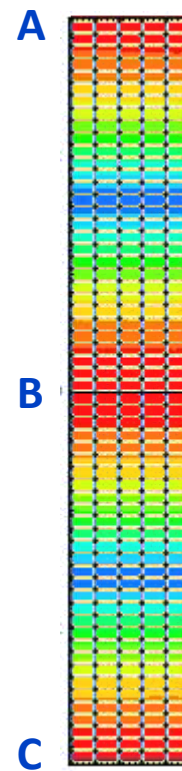
Warped Ring



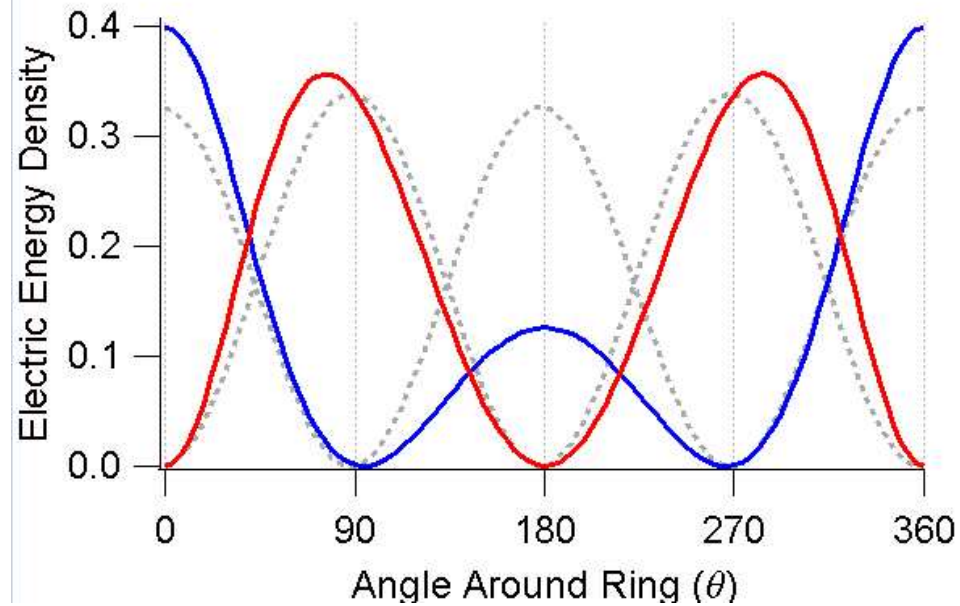
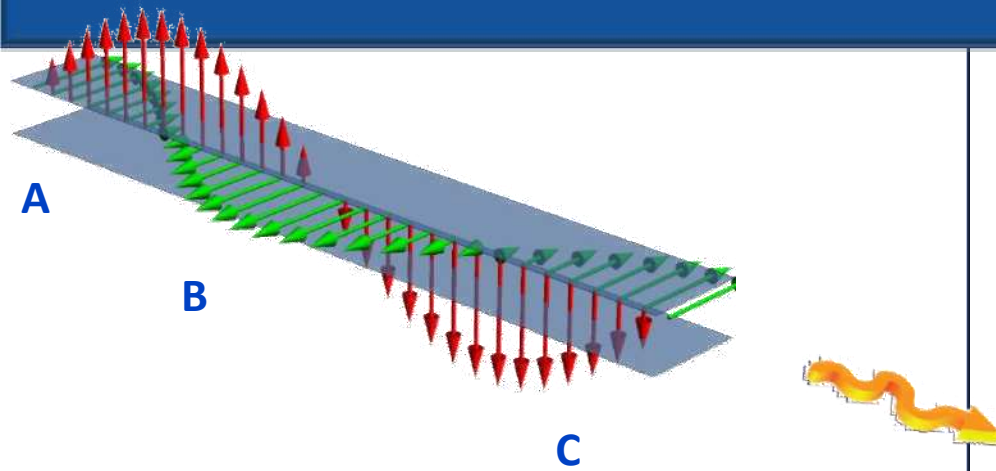
Top View ($|\vec{E}|$)

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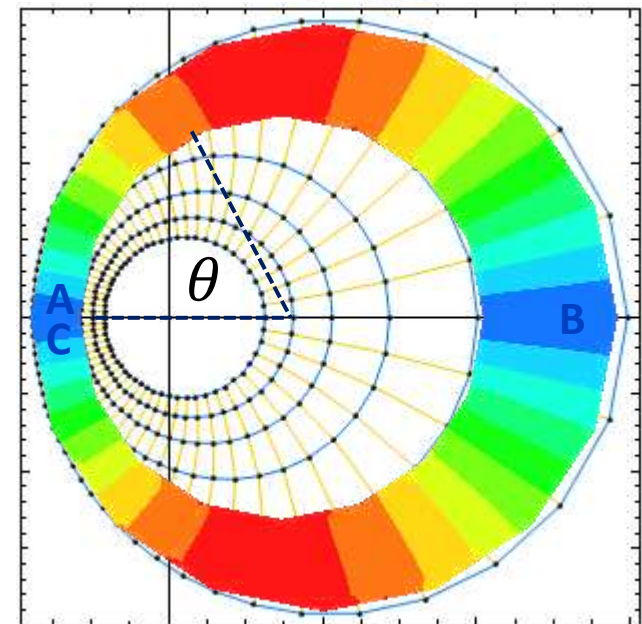
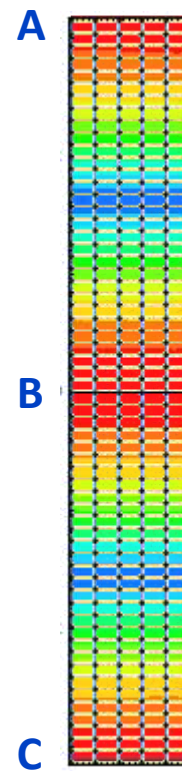
Warped Ring



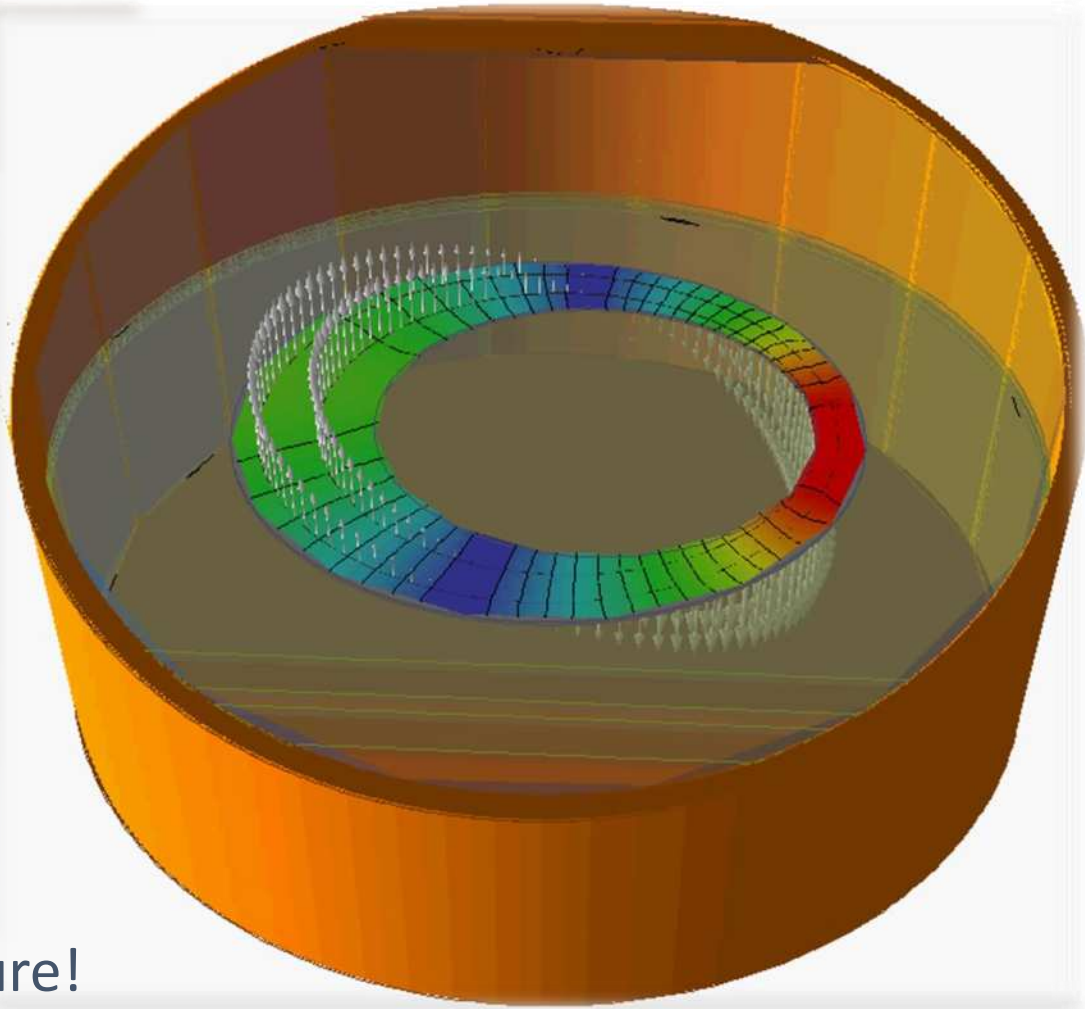
Top View ($|\vec{E}|$)

z-plane $\longrightarrow$ w-plane

$$w = \frac{e^z}{1 - s e^z}$$



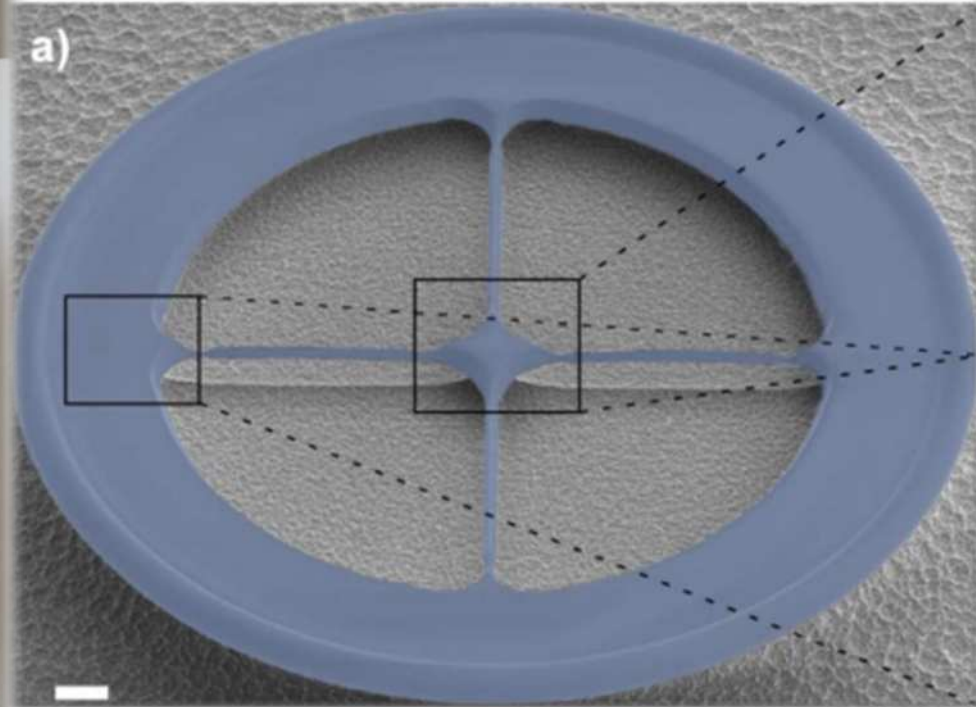
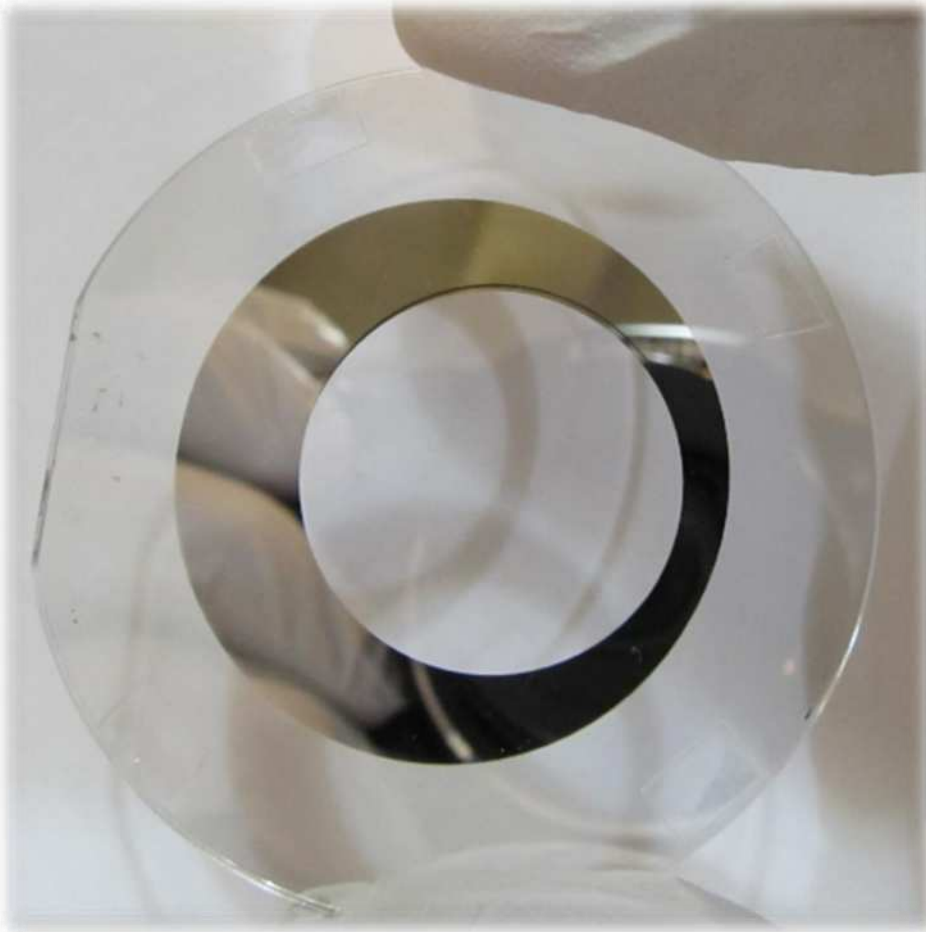
Towards 'Wafer-Scalable' cQED architecture



There are **2 rings** in this picture!

Whispering Gallery Modes

Similar to optical whispering gallery mode resonators



EPFL, *Nature Photonics* **2**, 627 - 633 (2008)

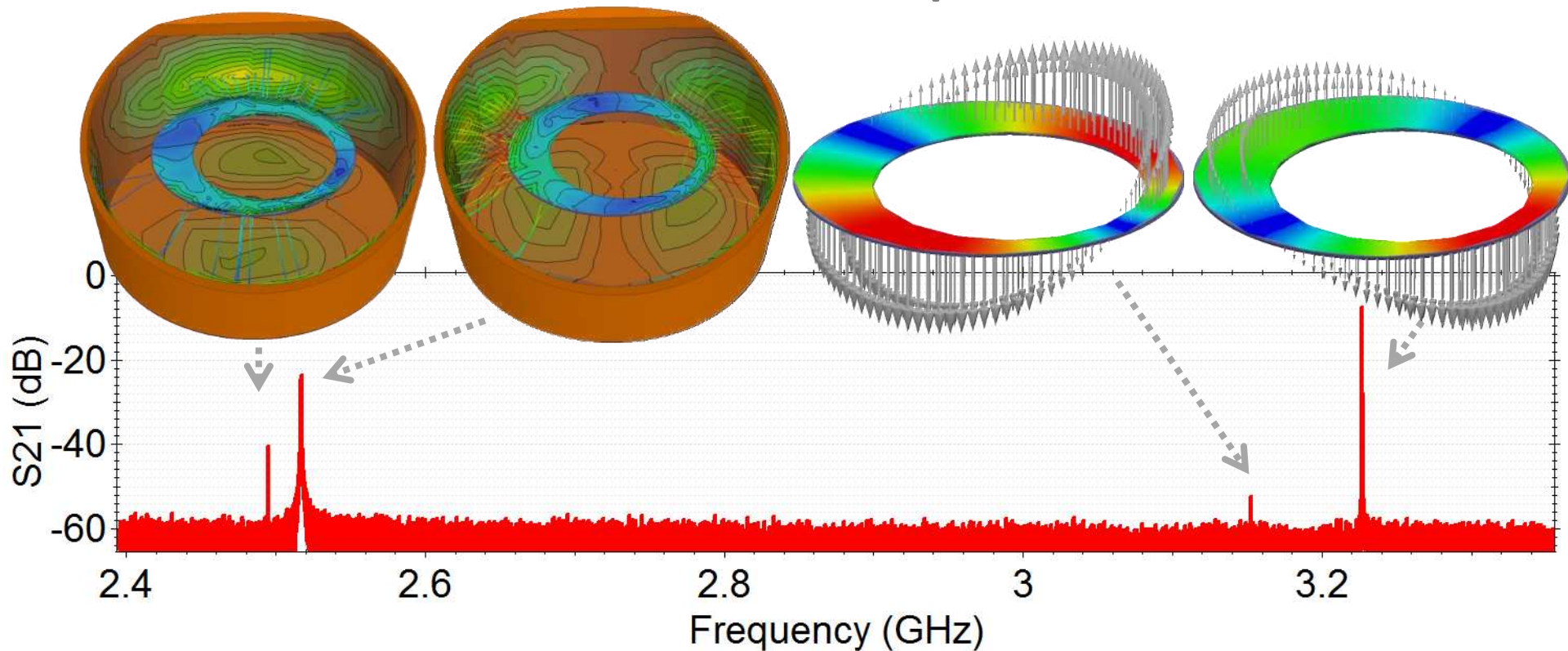
Identifying Modes (S_{21})

Common
Perpendicular

Common
Parallel

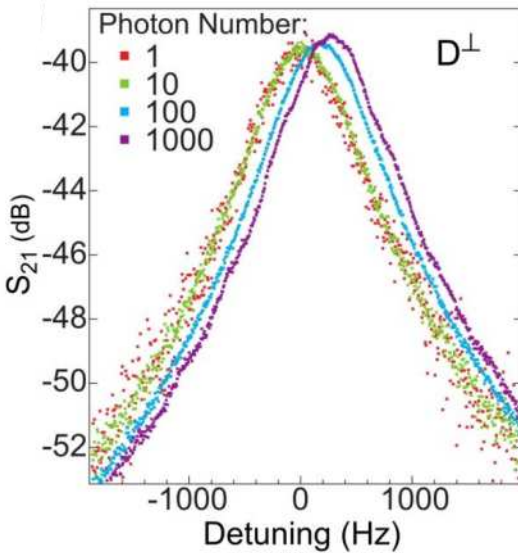
Differential
Perpendicular

Differential
Parallel



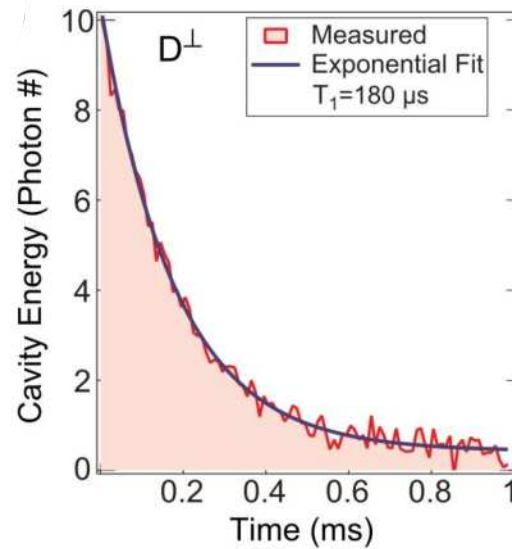
- Differential modes contain > 98% of mode energy in between the rings

Summary Superconducting WGMR Coherences

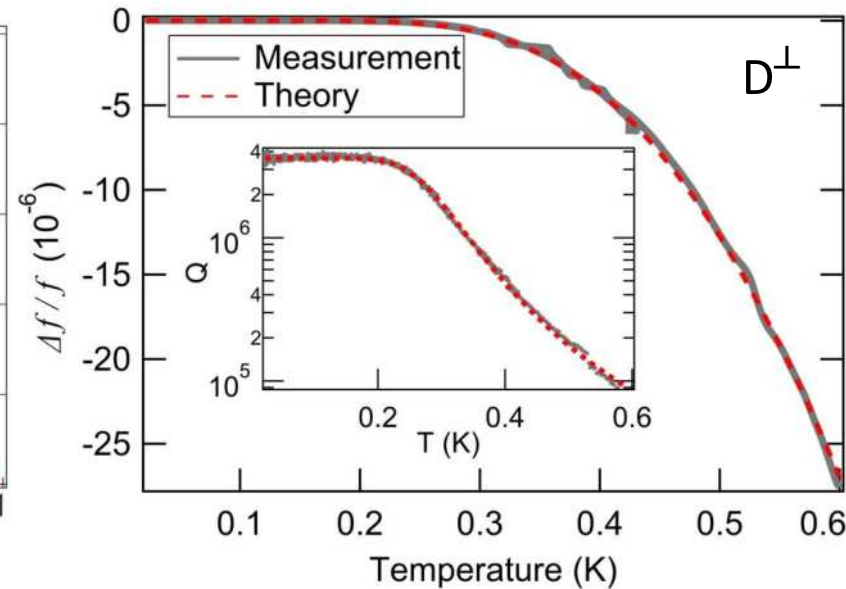


$$T_1 = 180 \mu\text{s}$$

$$Q_i = 3.4 \cdot 10^6$$



$$T_\phi \geq 1 \text{ ms}$$



Thin-film Al

$$R_s \leq 250 \text{ n}\Omega$$

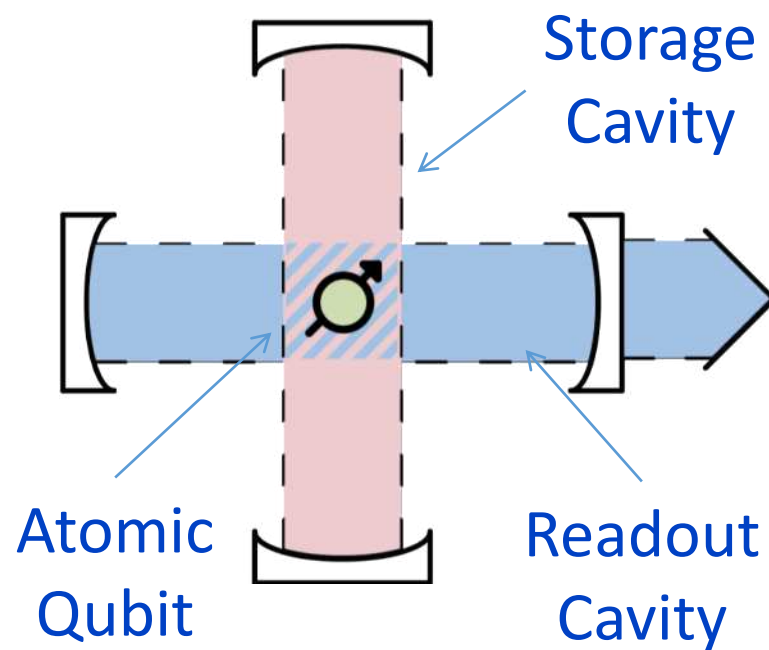
Sapphire

$$\tan \delta \leq 10^{-6}$$

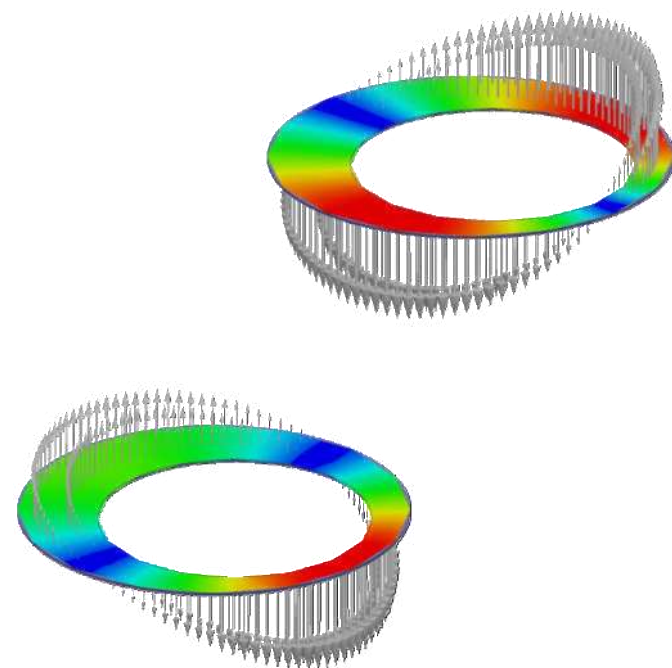
- Long-lived ($T_1 = 180 \mu\text{s}$) and coherent ($T_\phi \geq 1 \text{ ms}$) resonators
- We place an upper bound on thin-film Al resistance $R_s \leq 250 \text{ n}\Omega$ ($Q_s \geq 4,800$) at the single photon level
- We place an upper bound on the single photon loss tangent of sapphire $\tan \delta \leq 10^{-6}$

Qubit coupling to superconducting whispering gallery mode resonator

Cavity QED Equivalent

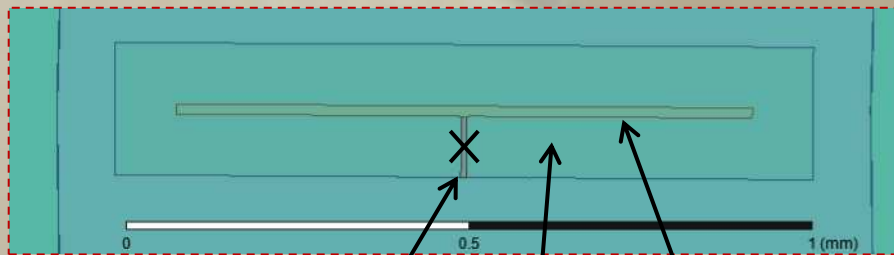


Ring Resonator





QUBIT CLOSE UP



Junction
Dielectric
(sapphire)

Metal
(aluminum)

Device Hamiltonian

In terms of the phase through the junction (φ), the system is described by:

$$H = \omega_r a^\dagger a + \omega_q b^\dagger b + \omega_m c^\dagger c - E_J [\cos(\varphi) - \frac{\varphi^2}{2}]$$

$$\varphi = \varphi_r (a^\dagger + a) + \varphi_q (b^\dagger + b) + \varphi_m (c^\dagger + c)$$

$\varphi_r, \varphi_q, \varphi_m$ refer to the zero point fluctuations of the junction phase in their respective mode.

These 6 parameters can be efficiently calculated with large precision $O(1\%)$.
The relevant lowest order non-linear (NL) pieces of the Hamiltonian then follow:

$$H_{NL} = \frac{1}{2} \alpha_r (a^\dagger a)^2 + \frac{1}{2} \alpha_q (b^\dagger b)^2 + \frac{1}{2} \alpha_m (c^\dagger c)^2$$

Anharmonicity (self-Kerr):

$$\alpha_r, \alpha_q, \alpha_m$$

$$+ \chi_{qr} a^\dagger a b^\dagger b + \chi_{qm} c^\dagger c b^\dagger b$$

Cross-Kerr

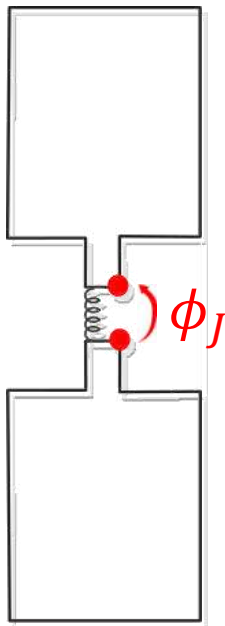
$$\chi_{qr}, \chi_{qm}$$

For instance:

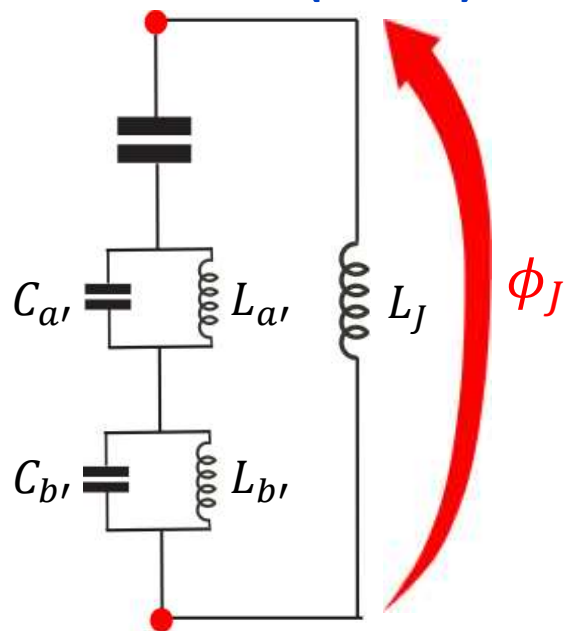
$$\alpha_r = \frac{1}{2} E_J \phi_r^4$$

BBQ: from device to circuit to Hamiltonian

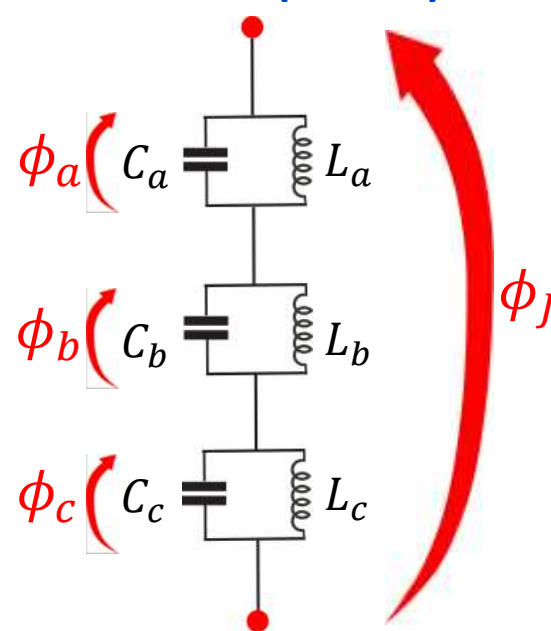
Physical (Linear)



Undressed (Linear)



Dressed (Linear)



Can we reformulate BBQ in terms of energy participation ratios?



$$\begin{aligned}\hat{\phi}_J &= \hat{\phi}_a + \hat{\phi}_b + \dots \\ &= \phi_a^{\text{ZPF}} (a + a^\dagger) + \phi_b^{\text{ZPF}} (b + b^\dagger) + \dots\end{aligned}$$

$$\phi_a^{\text{ZPF}} = \sqrt{\frac{\hbar}{2} Z_a^{\text{eff}}} = \sqrt{\frac{\hbar}{\omega_a \text{Im } Y'(\omega_a)}}$$

Nigg et al. PRL 108, 240502 (2012)

Energy Participation Ratios

from HFSS

$$\text{Energy in Mode } a = \overline{U_E} + (\overline{U_M} + \overline{U_K}) = 2 \overline{U_E}$$

$$p_J^a \equiv \frac{\text{Energy stored in Junction}}{\text{Energy stored in mode } a} = \frac{\overline{U_K}}{2 \overline{U_E}} = \frac{\overline{U_E - U_M}}{2 \overline{U_E}}$$

$$= \frac{\langle n_a | \frac{1}{2} E_J \hat{\phi}_J^2 | n_a \rangle}{\langle n_a | H | n_a \rangle} = \frac{E_J}{\hbar \omega_a} \phi_a^{\text{ZPF}^2}$$

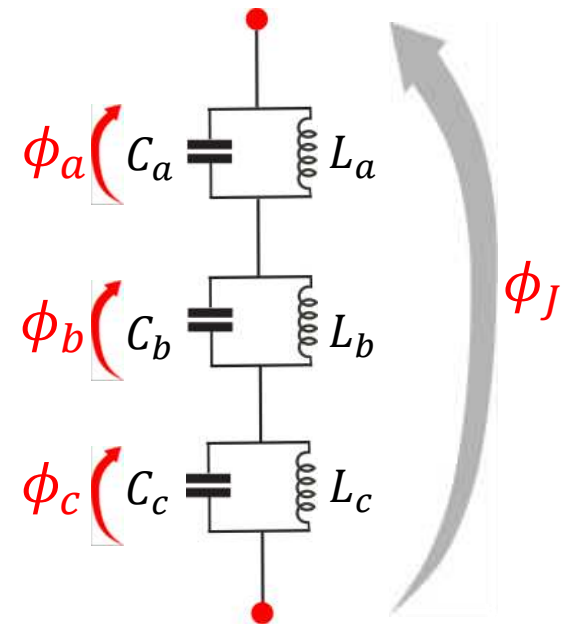
$$\phi_a^{\text{ZPF}^2} = p_J^a \frac{\hbar \omega_a}{E_J}$$

$$\alpha_a = \frac{1}{2} E_J \phi_a^{\text{ZPF}^4} = \frac{1}{2} \frac{(p_J^a \hbar \omega_a)^2}{E_J}$$

$$\chi_{ab} = p_J^a p_J^b \frac{\hbar \omega_a \hbar \omega_b}{E_J}$$



Dressed (Linear)



$$\hat{\phi}_J = \hat{\phi}_a + \hat{\phi}_b + \dots$$

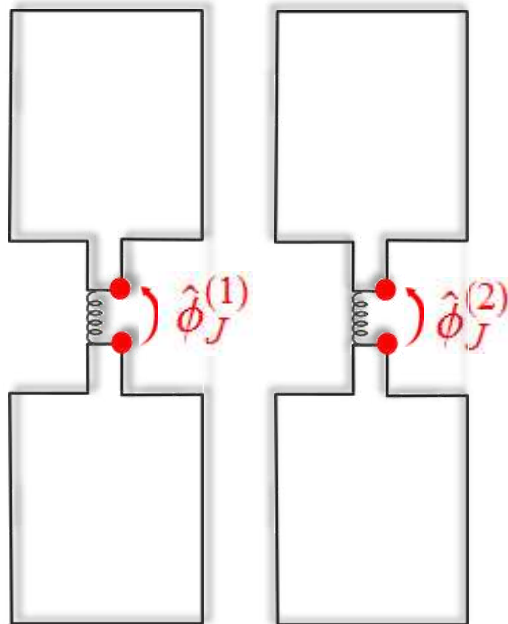
$$= \phi_a^{\text{ZPF}} (a + a^\dagger) + \phi_b^{\text{ZPF}} (b + b^\dagger) + \dots$$

$$L_a = 2 p_J^a L_J$$

$$C_a^{-1} = \omega_a^2 L_a$$

$$\sum_i p_J^{a_i} = \frac{1}{2}$$

Multi-Junction BBQ



$$H = \sum_i \omega_i a_i^\dagger a_i - \sum_p E_J^p \left[\cos(\hat{\phi}_p) - \frac{\hat{\phi}_p^2}{2} \right]$$

$$\hat{\phi}_J^{(1)} = \hat{\phi}_a^{(1)} + \hat{\phi}_b^{(1)} + \dots = \phi_a^{(1) \text{ ZPF}} (a + a^\dagger) + \phi_b^{(1) \text{ ZPF}} (b + b^\dagger) + \dots$$

$$\hat{\phi}_J^{(2)} = \hat{\phi}_a^{(2)} + \hat{\phi}_b^{(2)} + \dots = \phi_a^{(2) \text{ ZPF}} (a + a^\dagger) + \phi_b^{(2) \text{ ZPF}} (b + b^\dagger) + \dots$$

⋮

$$\hat{\phi}_J^{(p)} = \sum_{i=1}^M \left[\phi_a^{(p) \text{ ZPF}^2} = p_a^{(p)} \frac{\hbar \omega_a}{E_J^{(p)}} a_i^\dagger \right]$$

Nigg et al. PRL 108,
240502 (2012)

$$\text{Dissipationless} \Rightarrow \phi_i^{(k)} \in \mathbb{R}$$

Exploring all the consequences of the phase of ϕ^{ZPF} is subject matter for another talk.

Where is the Energy?



From “Energy BBQ” we can extract:

Hamiltonian

$$H = H(a^\dagger, a)$$

Interactions

$$\alpha_r, \alpha_q, \alpha_m, \chi_{qr}, \chi_{qm}, \dots$$

I/O rates

$$\kappa_r, \kappa_q, \kappa_m, \dots$$

Dissipation budget

$$R_s, \tan \delta, \dots$$

Advantages:

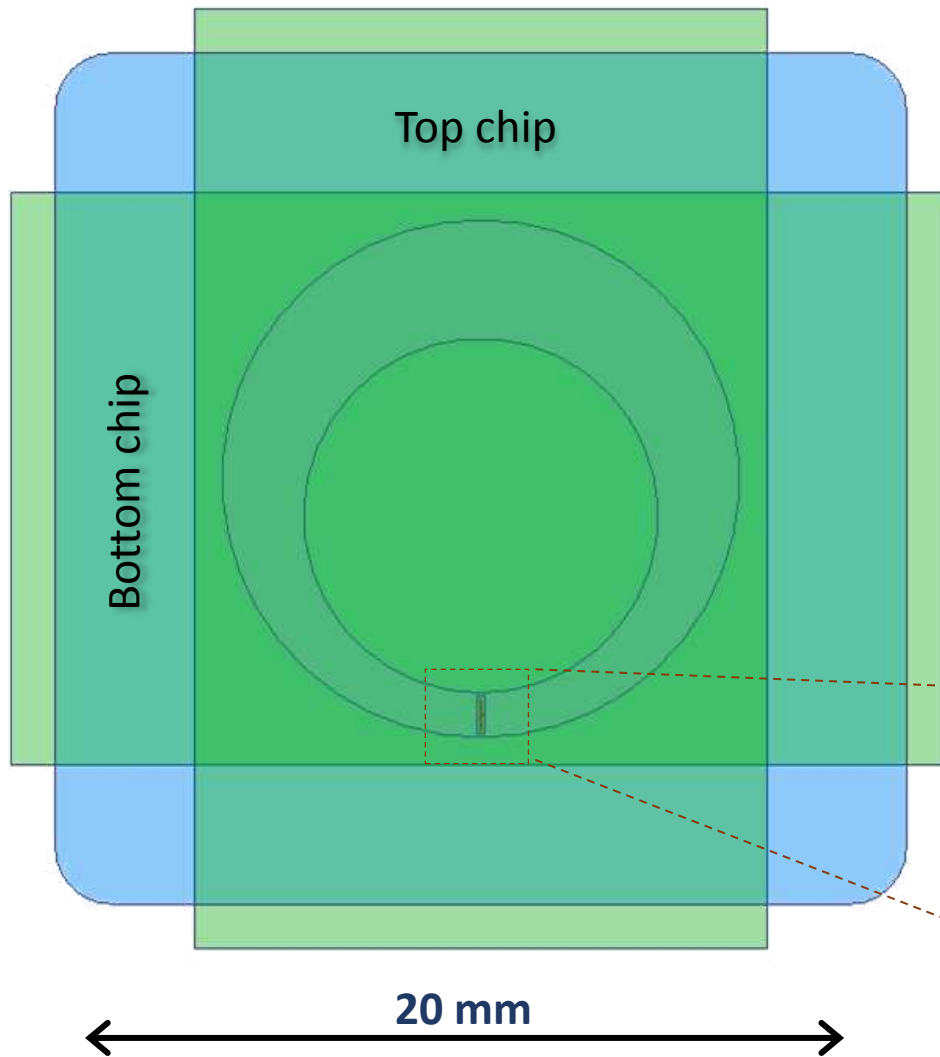
Conceptually simple framework

Only one HFSS simulation

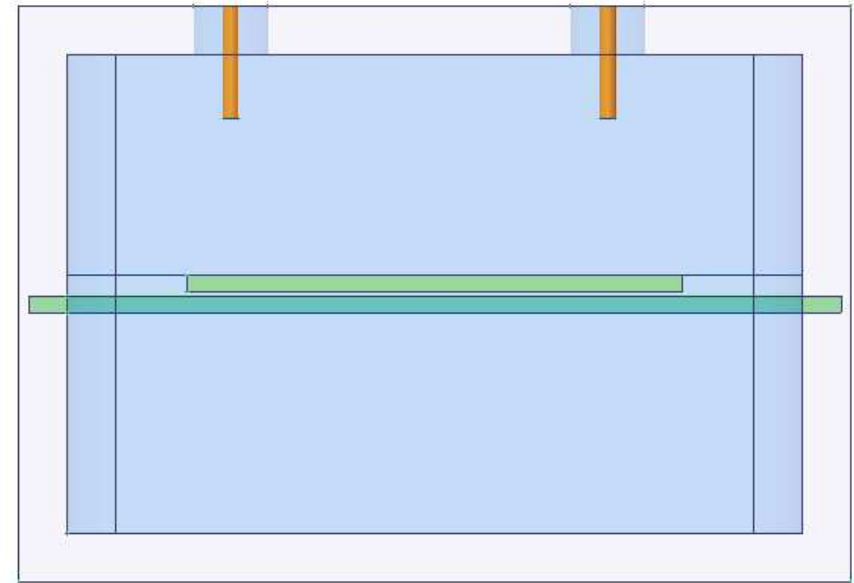
Predicts limitations - ex. bright state

Sample holder idea

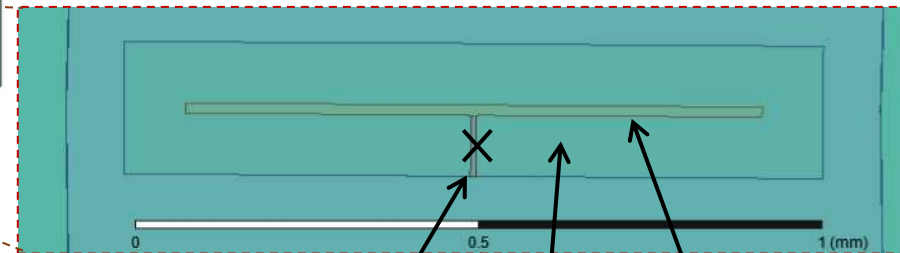
TOP VIEW



SIDE VIEW

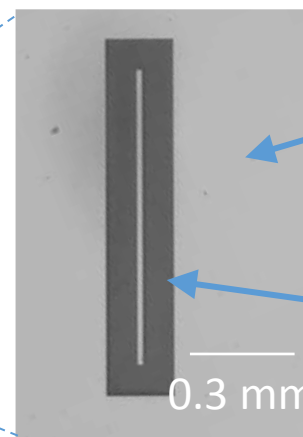


QUBIT CLOSE UP



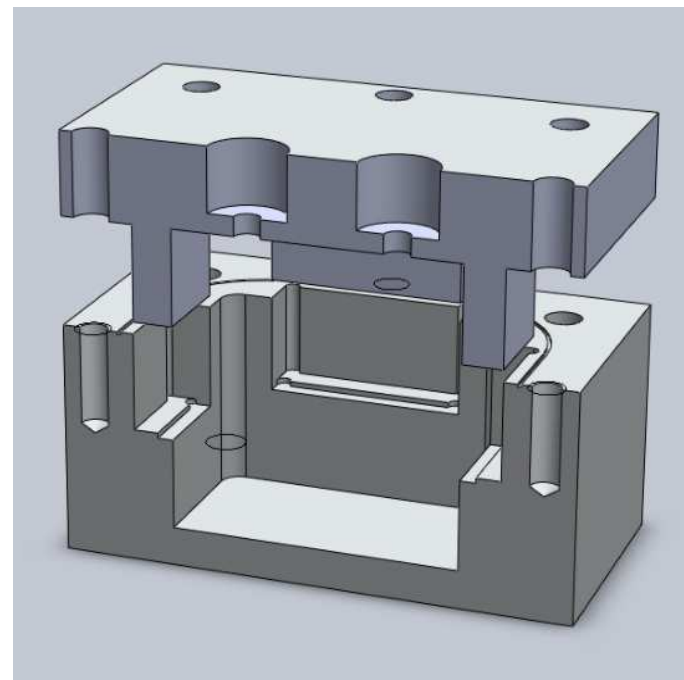
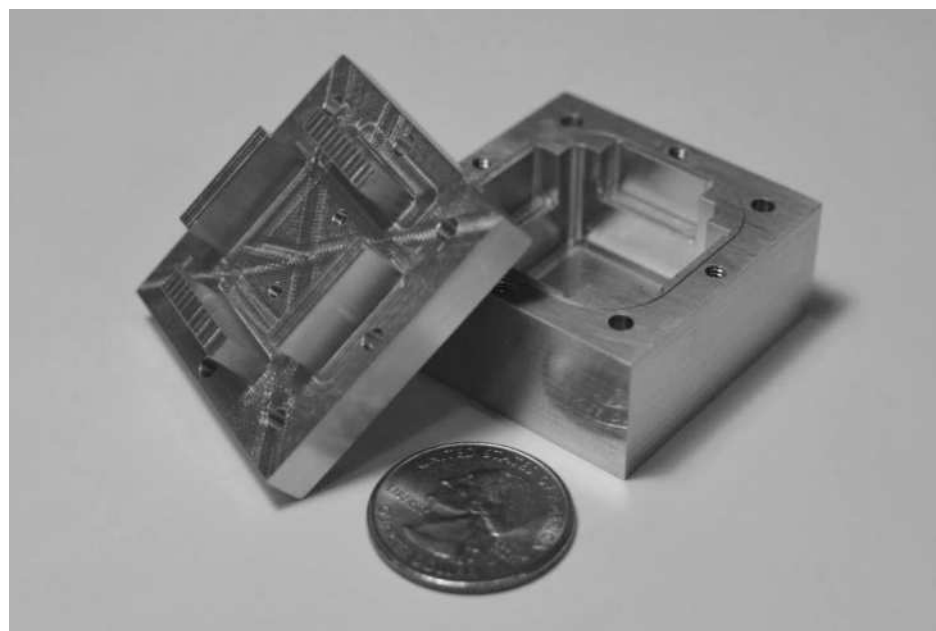
Junction Dielectric (sapphire) Metal (aluminum)

The device

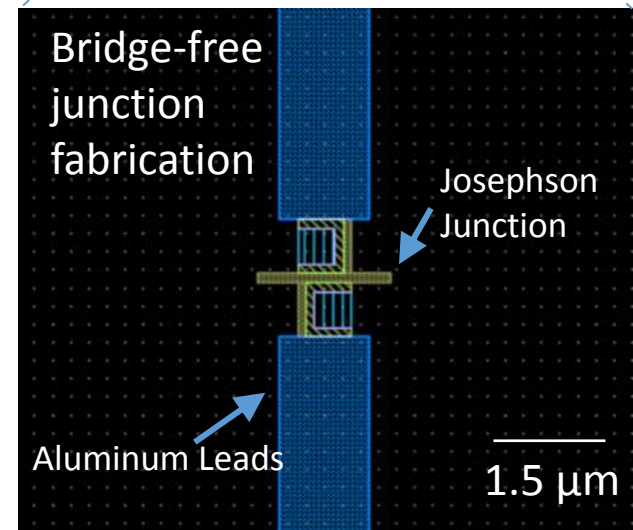
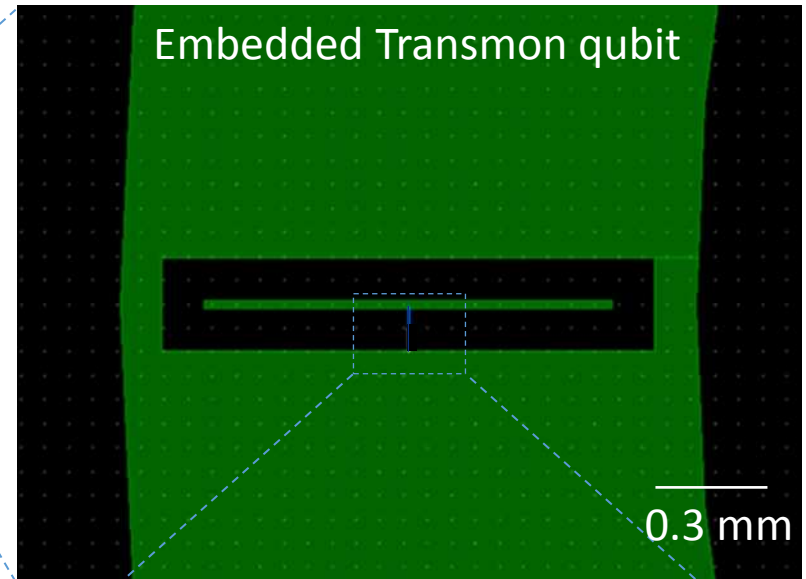
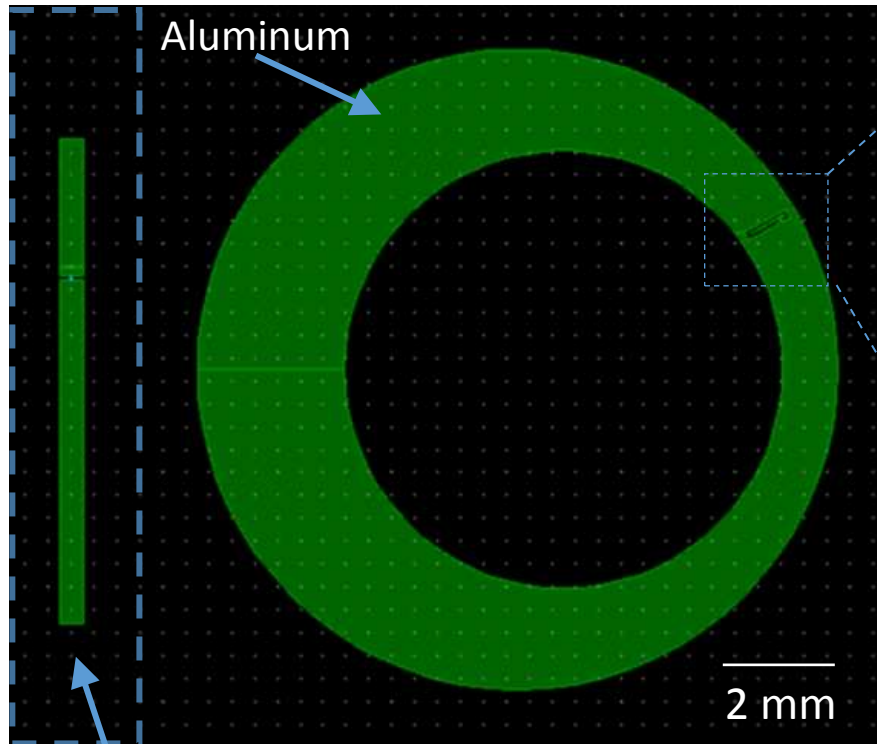


50 nm Aluminum

Al_2O_3 substrate,
430 μm thick



Fabrication: e-Beam Lithography



Zaki's Transmon (for size comparison)
100% JJ Survival Rate after dicing (6/6)
Bridge free fabrication (BFF)

F. Lecocq et al. Nanotechnology (2011)

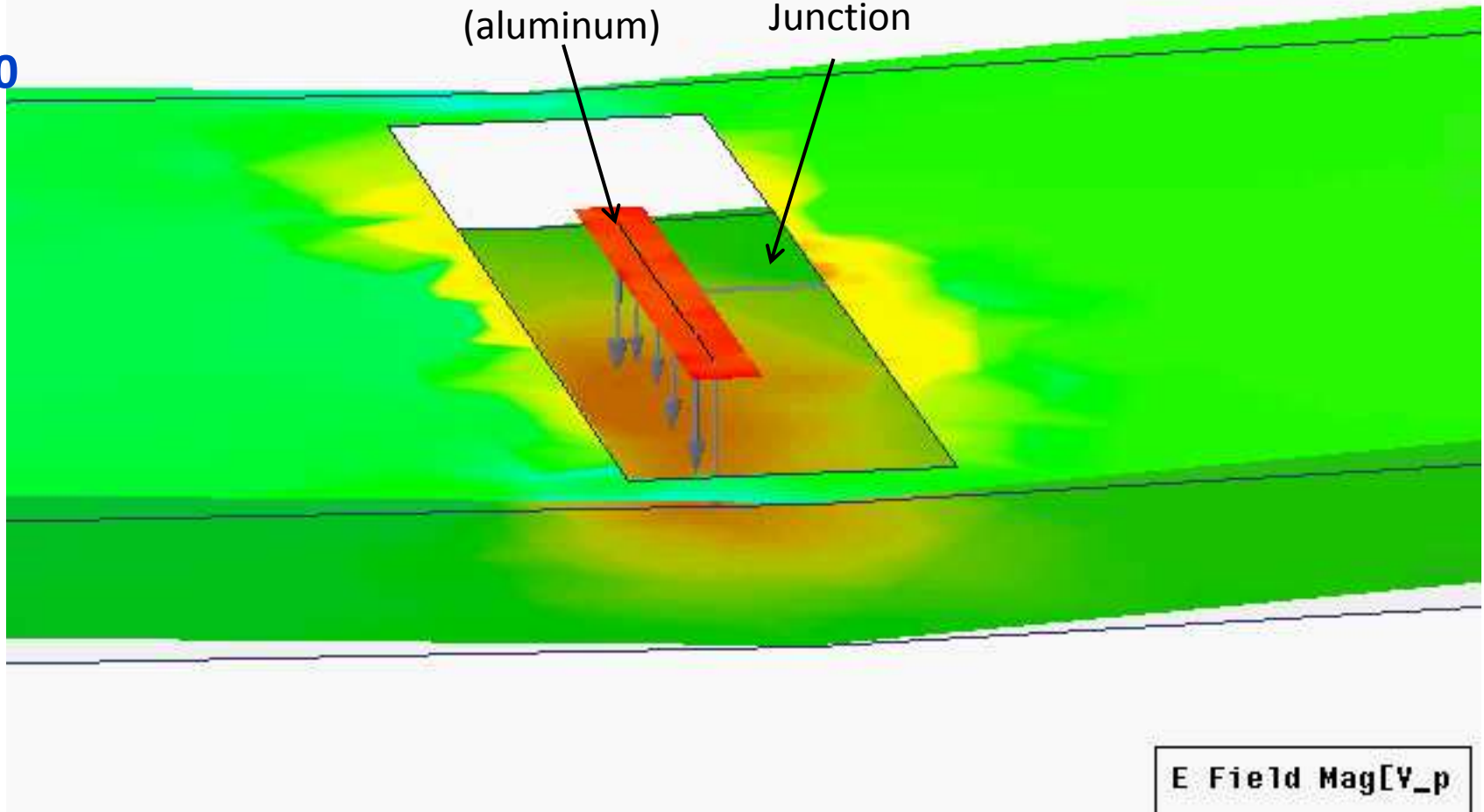
I.M. Pop et al. J. Vac. Sci. Technol. B (2012)

Qubit Mode

$\log|\vec{E}|$
Max
0

Metal
(aluminum)

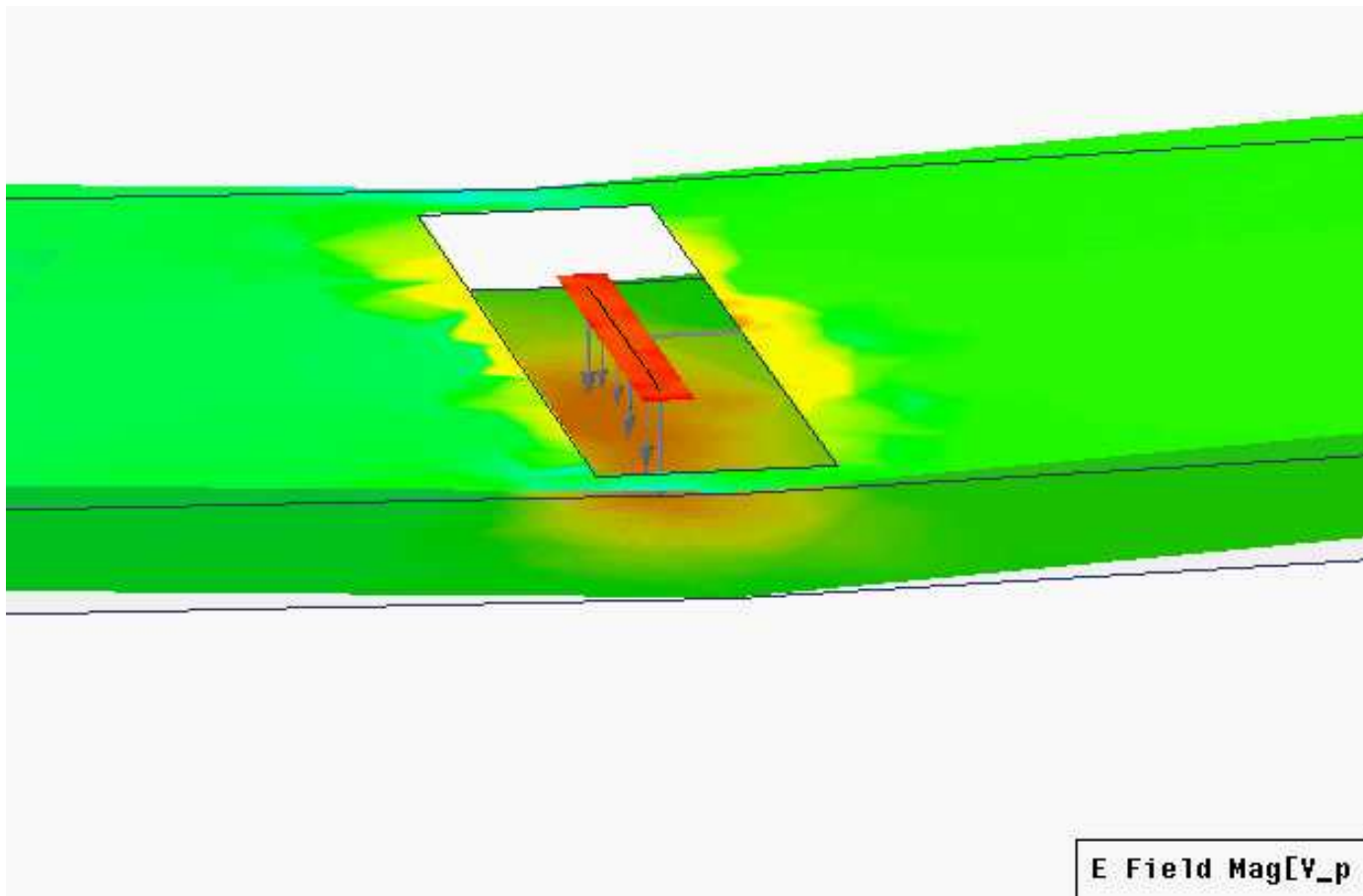
Junction



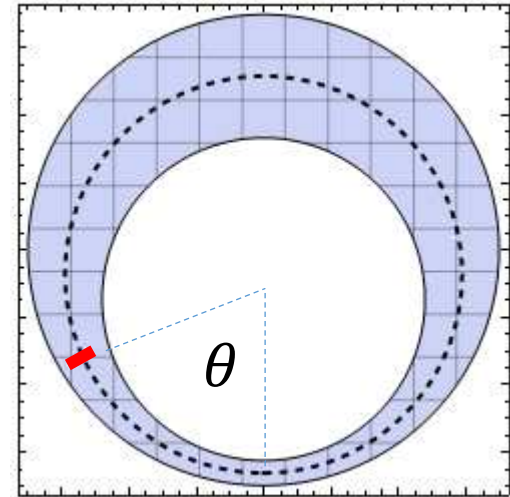
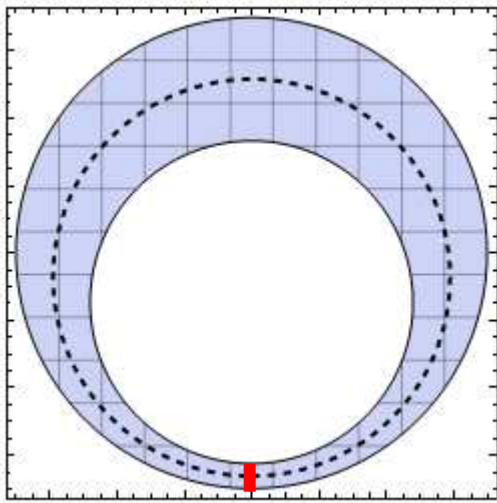
Qubit Mode

$\log|\vec{E}|$
Max
0

$|\vec{J}_{surf}|$
←

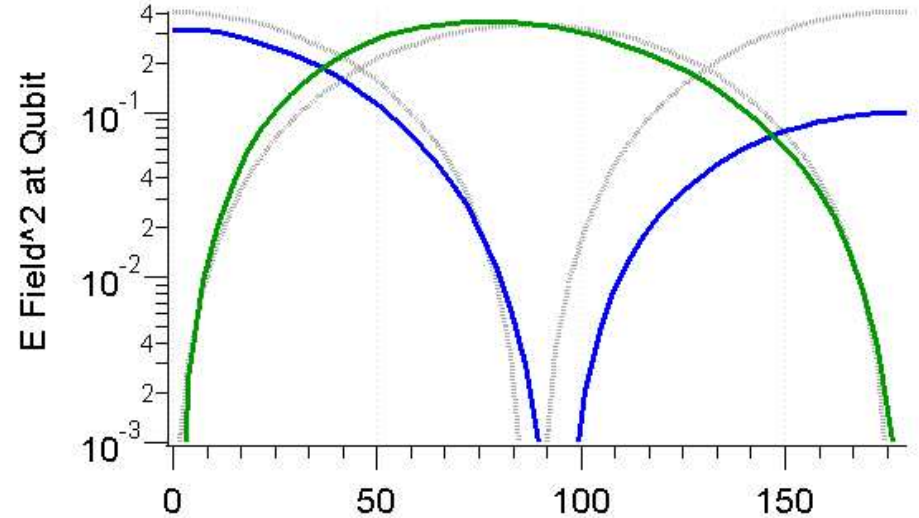
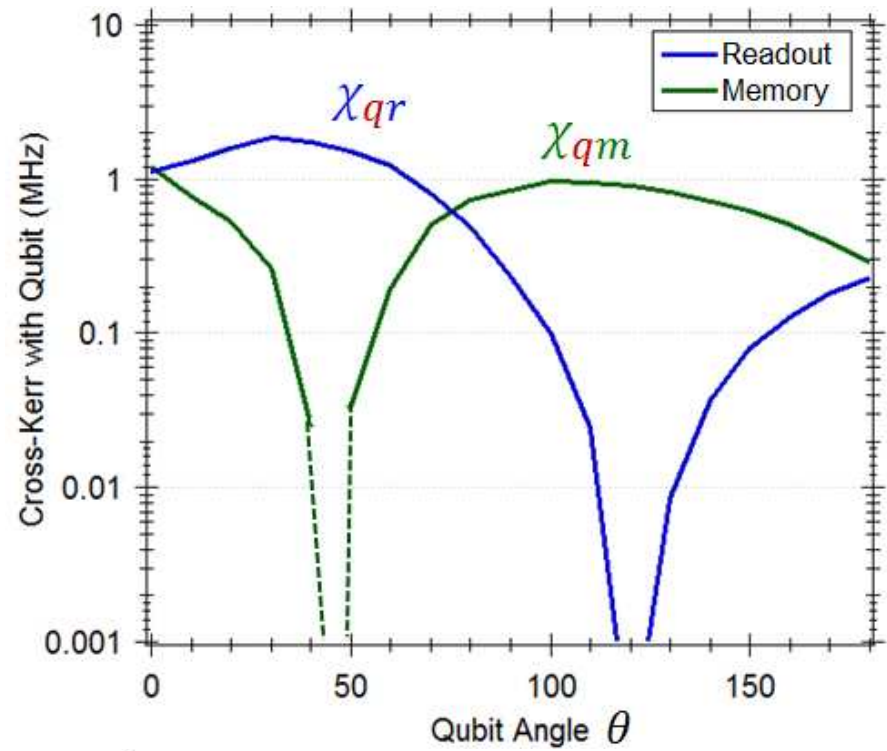
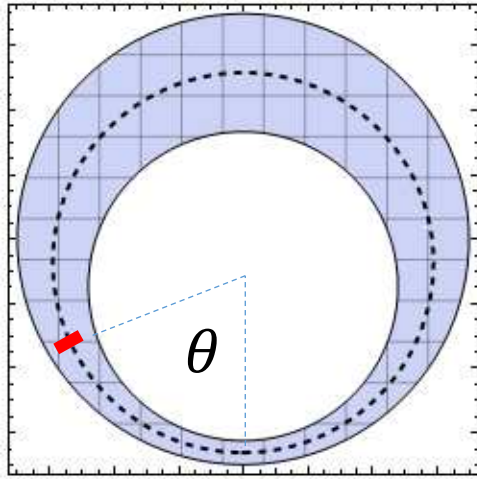


Key idea: can vary qubit angle around the ring



Qubit frequency and anharmonicity are independent of angle.

Tuning Cross-Kerr



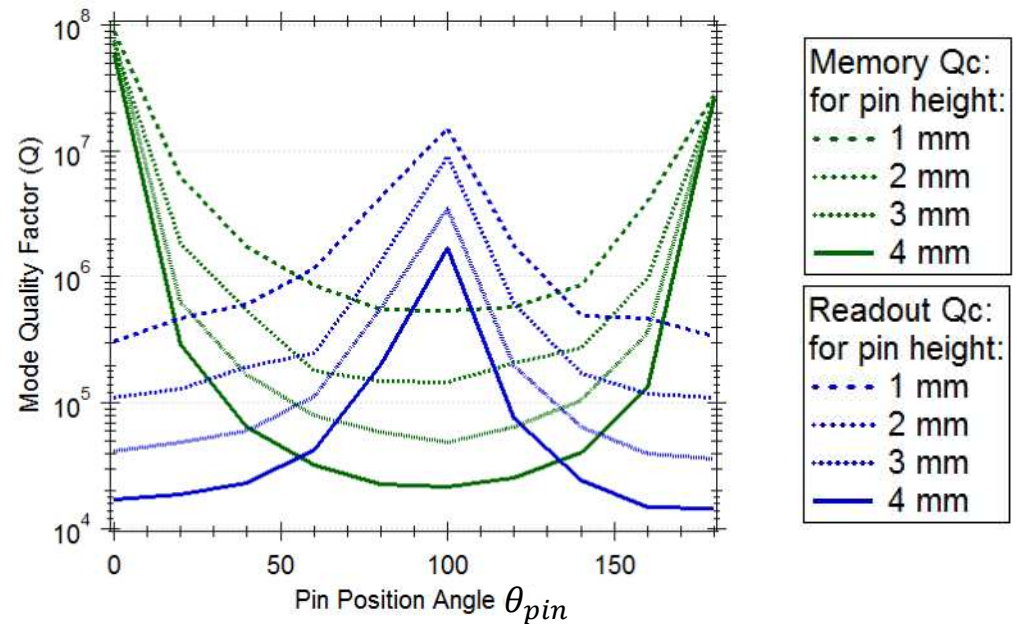
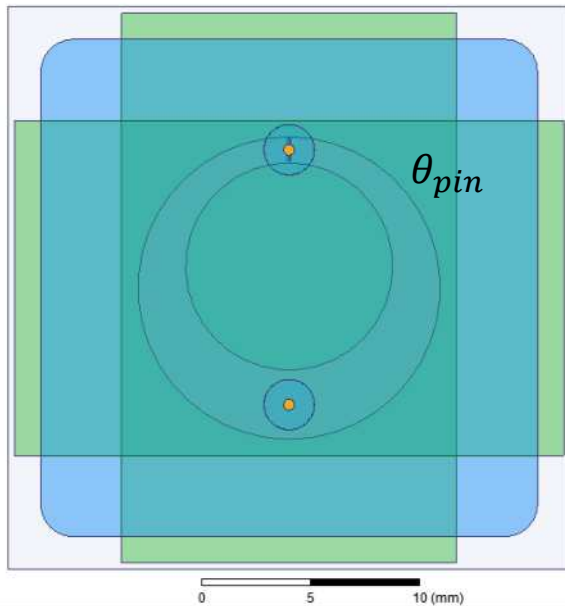
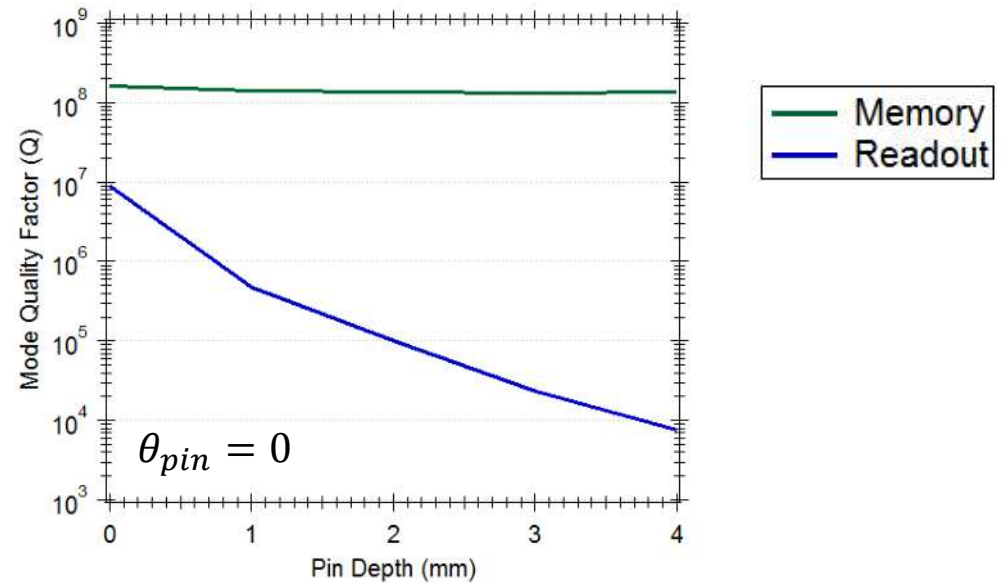
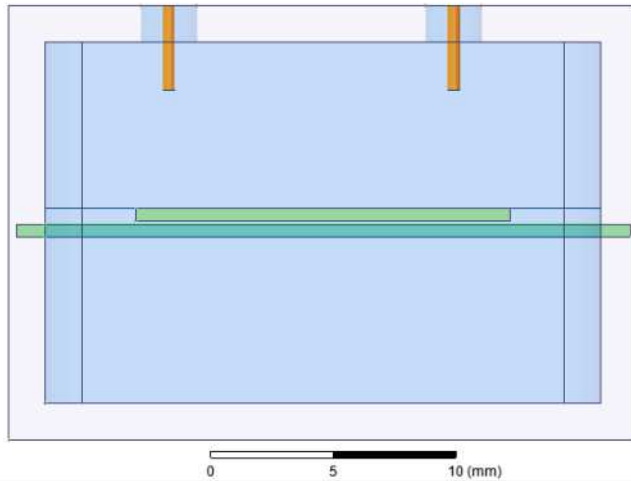
$$\omega_q / 2\pi = 6.1 \text{ GHz}$$

$$\omega_m / 2\pi = 7.2 \text{ GHz}$$

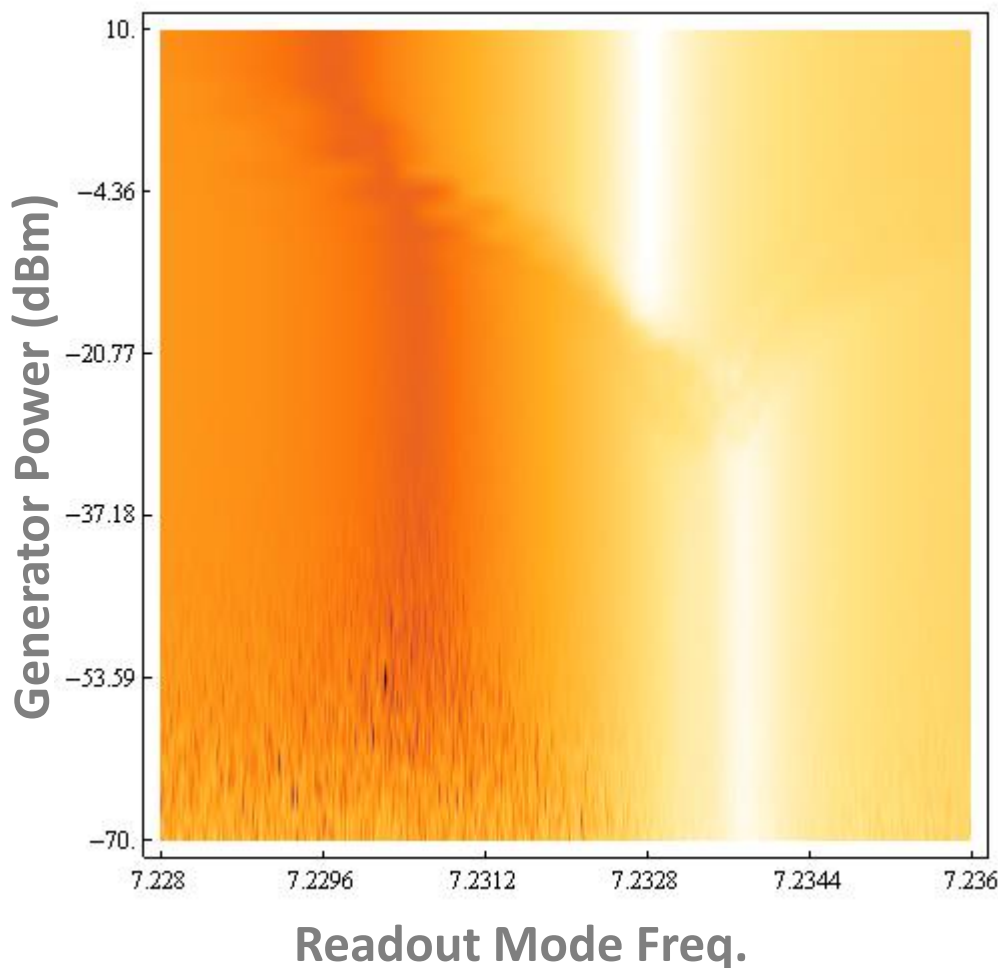
$$\omega_r / 2\pi = 7.6 \text{ GHz}$$

$$\alpha_q / 2\pi = 250 \text{ MHz}$$

Selective mode coupling



0th iteration preliminary results



Readout:

7.23 GHz (4%)

$Q \sim 16K$

Memory:

7.02 GHz (3%)

$T_m = 6.3 \mu s$

Qubit:

5.6 GHz (4%)

$T_1 = 15 \mu s$

Limited by alignment error induced coupling to pins.

Conclusion / Future Directions

- 2.5D resonators can be coherent at the level of parts per million
- “Energy BBQ” is fast, scalable and intuitive
- 2.5 planar qubit can achieve strong electric and magnetic coupling to ring resonators.
- Improve coherence
- Quantum optics with 2.5D cQED (multi-qubit)

Qlab

